

C# | Practical CTF

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Content

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For the complete documentation index, see [llms.txt](#). This page is also available as [Markdown](#).

Hello World[]()

The first step is creating a new project. With the `console` template for a simple CLI app, you can easily fill an empty directory with the necessary files:

Copy

```
mkdir HelloWorld && cd HelloWorld
dotnet new console
```

You can find external packages in the [NuGet Gallery](#) , and then add them to your project:

Copy

```
dotnet add package Newtonsoft.Json
```

Finally, run the main `Program.cs` file:

Copy

```
dotnet run
```

Deserialization[]()

There are different ways to serialize objects in C#, which is the process of turning it into a string. Then, this string can be passed around through other channels and eventually be **deserialized** to receive an identical copy of the original object.

Creating arbitrary objects with fields is dangerous when this deserialized string is in the attacker's control. By abusing lax configuration, you can instantiate objects with special behavior to read/write files, or even achieve Remote Code Execution if the right gadgets are accessible.

Newtonsoft Json.NET[]()

The most common form on deserialization in the web is JSON. The [Json.NET](#) library is the most widely-used for turning some string from the user into an instance of a class. The fields on this class define the structure of the JSON, for example ([source](#)):

Copy

```
public class Account {
    public string Email { get; set; }
    public bool Active { get; set; }
    public DateTime CreatedDate { get; set; }
    public IList<string> Roles { get; set; }
}

string json = @"{
    'Email': 'james@example.com',
    'Active': true,
    'CreatedDate': '2013-01-20T00:00:00Z',
    'Roles': [
        'User',
        'Admin'
    ]
}";

Account account = JsonConvert.DeserializeObject<Account>(json);
Console.WriteLine(account.Email); // "james@example.com"
```

The above example is **secure**, because it only allows deserializing basic data types. It can be wrongly configured, however, to allow all classes instead, which may include dangerous ones we call "gadgets". This is possible if a `JsonSerializerSettings` is given as the 2nd argument with a `.TypeNameHandling` value other than `None`.

This enables a special `$type` key for each JSON object (also in nested properties) that can reference any loaded class, and set its fields. This is only possible for properties with the `Object` type because all gadgets will inherit from it:

You can easily generate a payload by *serializing* it first with the same library and classes, then send it to the target. Make sure to include `TypeNameHandling.All` to ensure any types are included and the target can resolve them. You should **structure your classes exactly the same** as the target because the `$type` key includes this information:

When ran with `dotnet run`, this will generate the object with payload first, and then serialize it into JSON ready to send to the target. The 2nd part will similar the target receiving the string, and deserializing it into a vulnerable type. You will see that the `set {}` method is called twice:

The syntax is pretty simple, so if you want to, you can even handcraft these payloads. The syntax for the `$type` key is `Path.To.Class, AssemblyName`, where the path to the class is follows the nested structure of namespaces and classes to your gadget.

For another example, see the writeup below:

[\[image: Logo\]](#)Nexus Void | Jorian Woltjerjorianwoltjer.com

Writeup including a custom Json.NET deserialization chain to execute commands

Json.NET is far from the only library allowing arbitrary objects to be deserialized. To get an overflow, see the table below to understand which library supports what features:

Name	Format	Additional requirements	Comments
BinaryFormatter	Binary	No	ISerializable gadgets
NetDataContractSerializer	XML	No	ISerializable gadgets
SoapFormatter	SOAP XML	No	ISerializable gadgets
DataContractSerializer	XML	Control of expected Type or <code>knownTypes</code> or weak <code>DataContractResolver</code>	Setters gadgets Some ISerializable gadgets
XmlSerializer	XML	Control of expected Type	Quite limited; does not work with interfaces
JavaScriptSerializer	JSON	Insecure <code>TypeResolver</code>	Setters gadgets
DataContractJsonSerializer	JSON	Control of expected Type or <code>knownTypes</code>	Setters gadgets Some ISerializable gadgets
ObjectStateFormatter	Text, Binary	No	Uses <code>BinaryFormatter</code> internally; TypeConverters gadgets
LosFormatter	Text, Binary	No	Uses <code>ObjectStateFormatter</code> internally
BinaryMessageFormatter	Binary	No	Uses <code>BinaryFormatter</code> internally
XmlMessageFormatter	XML	Control of expected Type	Uses <code>XmlSerializer</code> internally

Table of serializers and what gadgets you can execute with them ([source](#))

Gadget Chains[]()

You'll be very lucky if you have the source code of your target application, and find a single setter in there that allows RCE. Instead, you should rely on chains of gadgets, often in widely-used libraries.

One small gadget can maybe call a function on another gadget, which grabs a property from a third gadget to ultimately use it in an unsafe way. It's an art to combine these in creative ways, and requires a good understanding of what's available and possible in the

codebase. The `ysoserial.net` tool collects such gadgets and can generate them with payloads at will:



[GitHub - pwntester/ysoserial.net: Deserialization payload generator for a variety of .NET for mattersGitHub](#)

Collection of gadget chains and generator for serialized input

To use it, select a gadget chain with `-g`, select the Formatter with `-f` (eg. `Json.Net`). Most gadgets will achieve RCE, and with the `-c` argument you can customize the final shell command it executes.

If the target loads the `PresentationFramework` assembly and you cause it to insecurely deserialize the above payload, the `calc.exe` command will be executed. If the conditions on the target are unknown, you should try many different known chains until one works.

Finding Gadgets[]()

To find your own gadgets, you should look for code that you are able to trigger during deserialization. These are `get {}` and `set {}` methods as mentioned above, but the **constructor will also be called**. You can pass named arguments to the constructor by your key names, for example:

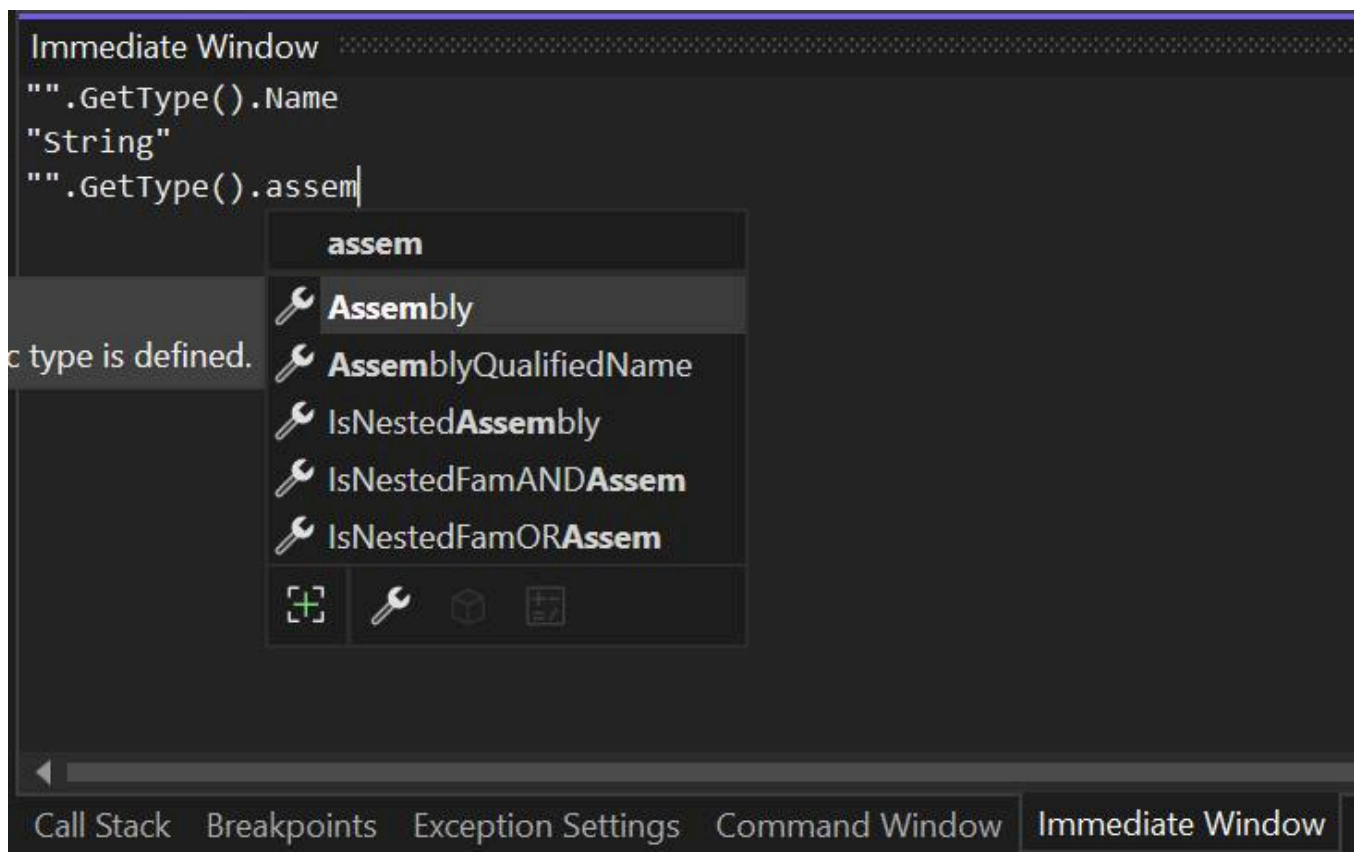
Some gadgets will call methods on your arguments, such as the `HashMap` calling `.hashCode()` to turn it into a unique integer. This means any vulnerable logic inside an object's `hashCode` implementation will also be callable if we just wrap in in a hashmap! Combining gadgets in chains like this is the standard way to find exploits.

Reflection[]()

Like many languages, C# has ways to interact with the type system at runtime through Reflection. This is useful in exploits when you can execute some limited C# code, or an interpreter of another language while having some interoperability. In such cases, you can often access properties and call methods on objects, and using Reflection, that can lead to RCE.

This is mainly done with chaining built-in methods on various types. All methods and attributes are well-documented on the Microsoft site, for example, the [Assembly class](#).

[Visual Studio](#) is the most featureful editor for C#. Something useful to us is when debugging any application, you can use the *Immediate Window* to quickly evaluate some small bits of code and get correct auto-completion. This makes it easier to explore your options.



Auto-complete feature and getting immediate results in Visual Studio

We'll go through an example of **ClearScript**, a JavaScript interpreter that [used to have an issue](#) allowing access to Reflection (and can still be configured to do so via `AllowReflection=true`).

Your first goal should be accessing the main `Assembly`, which you can get from a `Type` as `.Assembly`. To always get the main assembly, you can get the type of a type, which will always be the built-in type. In the example below, `Helper` was a C# object passed into the sandboxed context. We can use it to get a reference to the assembly:

We will now use its `Load(String)` method to import a built-in assembly that allows executing shell commands: `System.Diagnostics.Process`. We can get access to `MethodInfo` as a variable, and to call it, we'll use `Invoke(Object, Object[])` where the 2nd argument is an array representing the arguments passed to the method.

To create an array, in some cases, the simple `[]` syntax isn't possible. Using more methods, however, we can construct one out of thin air. We'll construct a new variable of type `List<String>` which has an `Add()` method. To do so, we need to pass `Assembly.CreateInstance()` a stringified version of the type, which we can get as follows:

Finally, to convert this mutable `List` into a `String[]`, we'll use its `ToArray()` method:

With this new `Process` assembly, we can prepare the arguments for its `Start(String, String)` method which takes the command to execute as its 1st argument, and the arguments (split by space) as shell arguments into the 2nd argument. If we list all the methods, this happens to be the 70th, and we can invoke it similar to before:

This should save the output of `id` into `/tmp/pwned`.

Similarly, the ****NVelocity**** templating framework can call arbitrary methods on C# objects, and thus is vulnerable to this Reflection abuse to reach RCE:

Finally, below is another exploit for the same framework that uses some different methods create a `ProcessStartInfo` and also return its output in the template content:

LINQ Injection[]()

Language Integrated Query (LINQ) is a Microsoft library for C# used to query objects similar to SQL syntax. It does, however, support C# syntax with function calls embedded inside the syntax, such as:

The above inserts user input from `showProducts.name` into the `Where()` call, which without sanitization allows an attacker to escape the `"` (double quote) and rewrite the query. For example:

```
-  
X") || 1==1 || "" == ("X: Shows all products
```

```
-  
X") || 1==2 || "" == ("X: Empty array
```

Version < 1.3.0 RCE[]()

The following Github Repository and accompanying article explain how to exploit such an injection for consistent Remote Code Execution.



[GitHub - Tris0n/CVE-2023-32571-POC](#)[GitHub](#)

Proof of Concept of the RCE

[\[image: Logo\]](#) Dynamic Linq Injection Remote Code Execution Vulnerability (CVE-2023-32571)[www.nccgroup.com](#)

Explanation and technical details of how it was found

Latest version property access[]()

The [patch](#) **only restricts method calling to predefined types**. This means that methods on Strings, Arrays, etc. will work, but methods on custom types will not. It is still possible to run methods on custom types that are inherited from allowed classes, and it is still possible to access any properties.

`"".GetType().Module.Assembly` still works to get the *Standard Module*.

`GetType().Module.Assembly` gets the module of the object passed into the `Where()` function, often custom code.

By chaining more properties and using `ToArray()` on enumerables, it is possible to enumerate all classes, attributes, properties and methods in a module. The following script implements this using binary search and requires a `test()` function that injects in such a way that you can evaluate a condition.

Example output looks like this (note that some magic members are also added, these can be ignored):

Filter Bypasses[]()

-

Any method call like `.GetType()` can be obfuscated as `.@GetType()`

-

Whitespace also works, eg. `. GetType()`

[Previous](#)[Java](#) [Next](#)[Assembly](#)

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